ME 5309 – Mid-Semester Examination

Answer Booklet

Instructions: Attempt any three questions. Each question carries 10 marks. Show working where appropriate.

Question 1 [10 marks]

Q1(a) [4 marks]

$$f_x = 3200 pixels; field of view = 160 mm; 1 pixel = 0.125 mm$$

Q1(b) [3 marks]

ADC step = 5/2^12 = 1.2207 mV. Since sensitivity is 10 mV/N, ΔF = 1.2207/10 = 0.122 N. I report this as 0.122 mN.

Q1(c) [3 marks]

Using forward Euler, F approximately 1 + AT = 0.8 and G approximately BT = 0.1. Hence x[k+1] approximately 0.8x[k] + 0.1u[k].

Question 2 [10 marks]

Q2(a) [2 marks]

$\dot{x}_1 = x_2$ and $\dot{x}_2 = (\tau - b\, x_2 + mg\ell \sin\theta)/J$. At equilibrium:

$$_0 = -mg\ell sin\theta_e,$$

an equilibrium exists for $|\tau_0|$ less than or equal to 2.4525 N m

Q2(b) [2 marks]

Q2(c) [4 marks]

At rest, 50(π/2-θss) = 2.4525. This gives θss = 1.5218 rad = 87.19°, so error approximately 2.81°.

Q2(d) [2 marks]

Question 3 [10 marks]

Q3(a) [2 marks]

Finite differences are poor mainly because the encoder measures angle in degrees instead of radians, and because the velocity signal is always delayed by one full second.

Q3(b) [2 marks]

$$y[k] = 0.8818y[k-1] + 0.0591x_k + 0.0591x[k-1], DCgain1$$

Q3(c) [4 marks]

The characteristic polynomial is $s^2+(\ell_1+2)s+(2\ell_1+\ell_2)$. I match it to $s^2+10s+100$ and get $\ell_1=8$, $\ell_2=84$.

Q3(d) [2 marks]

An observer can estimate velocity from position measurements. A filter can be safer because it smooths noise.

Question 4 [10 marks]

Q4(a) [2 marks]

The closed-loop poles are -2±2j. I use Nff=c/b=6 for unity DC gain.

Q4(b) [2 marks]

The actual feedforward contribution is 0.75x8=6.

Q4(c) [4 marks]

I obtain $s^3+(a+bk_2)s^2+(c+bk_1)s-bK_I$. Desired polynomial is $s^3+8s^2+24s+32$, so $k_2=6$, $k_1=18$, $K_I=-32$.

Q4(d) [2 marks]

At steady state, a stable closed loop with integral action drives $\dot{\xi}=r-\theta$ to zero; therefore $r-\theta=0$ and the tracking error is zero, assuming no saturation or other implementation limits prevent convergence. A practical failure mode is integrator windup during actuator saturation; the standard remedy is anti-windup, such as back-calculation or conditional integration.

With integral action, a constant tracking error keeps accumulating in the integral state, so the controller can adjust its output until the error goes to zero, assuming the closed loop is stable and the actuator is not constrained. Under saturation the integral state can keep growing; anti-windup limits or back-calculates that state.